

Lab 8

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INFT1200: Computer Forensics

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Questions

1. Amazon was an early provider of Web-based services that eventually developed into the cloud concept. True or False?

True – With Amazon Web Services, Amazon was one of the first web-based infrastructure services created in 2006 alongside S3 and EC2 to contribute to what cloud computing is today. Although Amazon did not invent all aspects of cloud computing but created an infrastructure that could provide cloud solutions at scale.

2. What are the three levels of cloud services defined by NIST?

- a. CRC, DRAM, and IMAP
- b. OpenStack, FROST, and management plane
- c. SaaS, PaaS, and IaaS**

Software as a Service provides software applications over the internet. Platform as a Service provides development solution applications. Infrastructure as a service provides computer infrastructure.

- d. Hybrid, private, and community clouds

3. What capabilities should a forensics tool have to acquire data from a cloud?

a. Identify and acquire data from the cloud.

Forensics tools use snapshots, APIs or remote collection to preserve chain of custody. These tools should allow you to locate assets, tag them, and perform evidence acquisition.

- b. Expand and contract data storage capabilities as needed for service changes.
- c. Circumvent firewalls to access cloud data.
- d. Examine virtual systems.

4. Commingled data isn't a concern when acquiring cloud data. True or False?

False – commingled data is a concern when acquiring data because the evidence can include unrelated data since it comes from multiple sources or users that are mixed together. The data should remain separated to avoid complicating investigations and creating a challenge in forensics.

5. A(n) _____ is a contract between a CSP and the customer that describes what services are being provided and at what level.

Cloud Service Agreement (CSA) outlines services provided by the cloud service provider (CSP) at the expected service levels.

6. Which of the following is a mechanism the ECPA describes for the government to get electronic information from a provider?

- a. Subpoenas with prior notice
- b. Temporary restraining orders

c. Search warrants

The Electronic Communications Privacy Act describes that the government can get electronic information from a provider using search warrants.

- d. Court orders

7. In which cloud service level can customers rent hardware and install whatever OSs and applications they need?

Infrastructure-as-a-Service (IaaS) is the cloud service level that customers can rent hardware and install any operating systems and applications they need through virtualized computing resources. Such resources include virtual machines, storage, networking capabilities, operating systems, and management of full applications.

8. What are the two states of encrypted data in a secure cloud?

- a. RC4 and RC5
- b. CRC-32 and UTF-16
- c. Homomorphic and AES

d. Data in motion and data at rest

In a secure cloud, the two states of encrypted data are data actively being transferred between locations and data that is not currently being accessed or moved.

9. Evidence of cloud access found on a smartphone usually means which cloud service level was in use?

- a. IaaS
- b. HaaS
- c. PaaS

d. SaaS

Software as a Service is most commonly associated with cloud access on a smartphone.

10. Which of the following cloud deployment methods typically offers no security?

- a. Hybrid cloud
- b. Public cloud
- c. Community cloud
- d. Private cloud**

11. The multitenancy nature of cloud environments means conflicts in privacy laws can occur. True or False?

True; multitenant environments can make it hard to differentiate data for each organization.

12. To see Google Drive synchronization files, you need a SQL viewer. True or False?

False. You can access Google Drive files directly through its interface.

13. A CSP's incident response team typically consists of which staff? List at least three positions.

System administrators, network administrators, and legal advisors.

14. The cloud services Dropbox, Google Drive, and OneDrive have Registry entries. True or False?

True

15. When should a temporary restraining order be requested for cloud environments?

- a. When cloud customers need immediate access to their data
- b. To enforce a court order
- c. When anti-forensics techniques are suspected
- d. When a search warrant requires seizing a CSP's hardware and software used by other parties not involved in the case**

16. Updates to the EU Data Protection Rules will affect how data is moved during an investigation regardless of location. True or False?

True

17. NIST document SP 500-322 defines more than 75 cloud services, including which of the following?

- a. Backup as a service**
- b. Security as a service
- c. Drupal as a service
- d. Intelligence as a service

18. Public cloud services such as Dropbox and OneDrive use what encryption applications?

AES 256-bit encryption